***Terminal screenshot (detail results are in folder /results)

```
PS C:\Users\niranjan patil\Desktop\sem5\ot\241070055_Niranjan_Patil__OT_lab8\optimize-anything> pip instal
ll numpy scipy matplotlib pytest pennylane scikit-learn coverage
Requirement already satisfied: numpy in c:\users\niranjan patil\anaconda3\lib\site-packages (2.3.5)
Requirement already satisfied: scipy in c:\users\niranjan patil\anaconda3\lib\site-packages (1.16.3)
Requirement already satisfied: matplotlib in c:\users\niranjan patil\anaconda3\lib\site-packages (3.10.6)
Requirement already satisfied: pytest in c:\users\niranjan patil\anaconda3\lib\site-packages (8.4.2)
Requirement already satisfied: pennylane in c:\users\niranjan patil\anaconda3\lib\site-packages (0.45.1)
Requirement already satisfied: scikit-learn in c:\users\niranjan patil\anaconda3\lib\site-packages (1.9.0
)
Collecting coverage
  Downloading coverage-7.17.0-cp313-cp313-win_amd64.whl.metadata (8.3 kB)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\niranjan patil\anaconda3\lib\site-packages (f
rom matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in c:\users\niranjan patil\anaconda3\lib\site-packages (from
matplotlib) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\niranjan patil\anaconda3\lib\site-packages (
from matplotlib) (4.60.1)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\niranjan patil\anaconda3\lib\site-packages (
from matplotlib) (1.4.9)
Requirement already satisfied: packaging>=20.0 in c:\users\niranjan patil\anaconda3\lib\site-packages (fr
om matplotlib) (25.0)
Requirement already satisfied: pillow>=8 in c:\users\niranjan patil\anaconda3\lib\site-packages (from mat
plotlib) (12.0.0)
Requirement already satisfied: pyparsing>=2.3.1 in c:\users\niranjan patil\anaconda3\lib\site-packages (f
rom matplotlib) (3.2.5)
Requirement already satisfied: python-dateutil>=2.7 in c:\users\niranjan patil\anaconda3\lib\site-package
s (from matplotlib) (2.9.0.post0)
Requirement already satisfied: colorama>=0.4 in c:\users\niranjan patil\anaconda3\lib\site-packages (from
 pytest) (0.4.6)
Requirement already satisfied: iniconfig>=1 in c:\users\niranjan patil\anaconda3\lib\site-packages (from
pytest) (2.1.0)
Requirement already satisfied: pluggy<2,>=1.5 in c:\users\niranjan patil\anaconda3\lib\site-packages (fro
```

```
PS C:\Users\niranjan patil\Desktop\sem5\ot\241070055_Niranjan_Patil__OT_lab8\optimize-anything\trackA> py
thon run.py
{
 "deterministic": {
  "convex_quadratic": {
   "GD": {
    "best_lr": 0.03,
    "final_f": 9.905198613333639e-09,
    "iters": 344,
    "n_lr_diverged": 2,
    "n_lr_converged": 2
   },
   "Momentum": {
    "best_lr": 0.01,
    "final_f": 2.989453870800249e-09,
    "iters": 167,
    "n_lr_diverged": 2,
    "n_lr_converged": 5
   },
   "NAG": {
    "best_lr": 0.03,
    "final_f": 8.734516478039205e-11,
    "iters": 87,
    "n_lr_diverged": 3,
    "n_lr_converged": 4
   },
   "AdaGrad": {
    "best_lr": 1.0,
    "final_f": 8.582096707950635e-09,
    "iters": 106,
    "n_lr_diverged": 0,
```

```
PS C:\Users\niranjan patil\Desktop\sem5\ot\241070055_Niranjan_Patil__OT_lab8\optimize-anything\trackB> py
thon run.py
=== B1. 0/1 Knapsack ===
  Greedy=147  DP=151  B&B=151  ILP=151
  Times: greedy=0.0005s  dp=0.0006s  bnb=0.0001s  ilp=0.0198s

=== B2. Weighted Interval Scheduling ===
  Greedy-EFT weight=25  DP weight=23  gap=-8.7%

=== B3. Max-Cut (7 nodes) ===
  Edges=11  Greedy=22  Brute-force=33
  SA 10 runs: mean=33.0  std=0.0

=== B4. QUBO / Ising formulation ===
  QUBO unconstrained: brute-force energy=-33.00  cut=33
  SA unconstrained 10 runs: mean=-33.00  std=0.00
  QUBO constrained (|S|=3, lam=10.0): brute-force energy=-123.00  cut=33
  Ising(h,J)+const = -33.0000  QUBO energy = -33.0000  (match OK)

Results saved to C:\Users\niranjan patil\Desktop\sem5\ot\241070055_Niranjan_Patil__OT_lab8\optimize-anyth
ing\trackB\..\results\trackB_results.json
```

```
PS C:\Users\niranjan patil\Desktop\sem5\ot\241070055_Niranjan_Patil__OT_lab8\optimize-anything\trackC> py
thon run.py
=== Track C — Quantum Optimization (PennyLane Simulator) ===
STATUS: SIMULATED on classical hardware (default.qubit / default.mixed)

--> Running C1: QAOA on Track B QUBO...
--> Running C2: VQE on 4-qubit Ising chain...
--> Running C3: Optimizer comparison (Parameter-Shift vs SPSA vs QNG)...
--> Running C4: PQC / QNN Binary Classifier...
--> Running C5: Finite-Shot Measurement Effect...
C:\Users\niranjan patil\anaconda3\Lib\site-packages\pennylane\devices\device_api.py:207: PennyLaneDeprec
tionWarning: Setting shots on device is deprecated. Please use the `set_shots` transform on the respecti
e QNode instead.
  warnings.warn(
--> Running C6: Hardware Noise Model Effect...
--> Running C7: Barren Plateaus (Gradient Variance Decay)...
  Qubits= 2: Var[grad]=1.992900e-01
  Qubits= 4: Var[grad]=8.415898e-02
  Qubits= 6: Var[grad]=4.751004e-02
  Qubits= 8: Var[grad]=1.850905e-02

[OK] Results saved to C:\Users\niranjan patil\Desktop\sem5\ot\241070055_Niranjan_Patil__OT_lab8\optimize-
anything\trackC\..\results\trackC_results.json
```

```
PS C:\Users\niranjan patil\Desktop\sem5\ot\241070055_Niranjan_Patil__OT_lab8\optimize-anything\trackD> p
thon run.py
  Greedy=30  Exact=30  ratio=1.0000  theory>=0.6321
  Diminishing returns: 0/500 violations (expected 0)

=== D2a/b. Makespan & Work Stealing ===
  LPT makespan=41  Random mean=49.1  LB=40
  Work stealing mean=41.9±0.7

=== D2c. Measured Speedup (multiprocessing) ===
  workers=1: t=0.473s  speedup=0.17x  eff=0.17
  workers=2: t=0.531s  speedup=0.15x  eff=0.07
  workers=4: t=0.622s  speedup=0.13x  eff=0.03

=== D3. FedAvg vs Centralized ===
  ls=1 rounds=5: FedAvg gap=4.7509  Central gap=3.4276
  ls=1 rounds=10: FedAvg gap=4.4469  Central gap=2.0384
  ls=1 rounds=20: FedAvg gap=3.5264  Central gap=0.9655
  ls=5 rounds=5: FedAvg gap=3.2836  Central gap=0.7098
  ls=5 rounds=10: FedAvg gap=1.8008  Central gap=0.1088
  ls=5 rounds=20: FedAvg gap=0.7131  Central gap=0.0086
  ls=10 rounds=5: FedAvg gap=1.9630  Central gap=0.1088
  ls=10 rounds=10: FedAvg gap=0.5541  Central gap=0.0086
  ls=10 rounds=20: FedAvg gap=0.0970  Central gap=0.0160

=== D4. Multi-task weighting ===
  equal: MSE=0.1450±0.0108  Acc=0.8994±0.0251
  mgda: MSE=2.2509±0.8106  Acc=0.9422±0.0117
  gradnorm: MSE=0.1865±0.0239  Acc=0.9372±0.0111

Results saved to C:\Users\niranjan patil\Desktop\sem5\ot\241070055_Niranjan_Patil__OT_lab8\optimize-anyt
ing\trackD\..\results\trackD_results.json
```

```
PS C:\Users\niranjan patil\Desktop\sem5\ot\241070055_Niranjan_Patil__OT_lab8\optimize-anything\trackE> cd..
PS C:\Users\niranjan patil\Desktop\sem5\ot\241070055_Niranjan_Patil__OT_lab8\optimize-anything> cd trackD
PS C:\Users\niranjan patil\Desktop\sem5\ot\241070055_Niranjan_Patil__OT_lab8\optimize-anything\trackD> python run_locality_sync.py
{
 "cpus": 4,
 "D5_data_locality": {
  "sum_rows_s": 0.0563149000518024,
  "sum_cols_s": 0.17610420007258654,
  "col_over_row_slowdown": 3.1271333148171006,
  "transpose_by_tile_s": {
   "8": 0.34686100017279387,
   "16": 0.12640060018748045,
   "32": 0.07054050033912063,
   "64": 0.05819079978391528,
   "128": 0.06209990009665489,
   "256": 0.05949210003018379,
   "1024": 0.05183310015127063,
   "3000": 0.06746549997478724
  },
  "transpose_best_tile": "1024"
 },
 "D6_sync_comm": {
  "nproc": 4,
  "serial_s": 0.005312999710440636,
  "lock_per_element_s": 1.5747382999397814,
  "local_partial_sums_s": 0.5083675999194384,
```

```
PS C:\Users\niranjan patil\Desktop\sem5\ot\241070055_Niranjan_Patil__OT_lab8\optimize-anything\trackE> py
thon run.py 20 40 60 100
{"n": 20, "T": 13, "edges": 148, "random": {"mean": 296.4, "std": 11.943198901466891, "feasible_runs": 10
, "time": 0.004636454582214356}, "greedy": {"obj": 244.0, "time": 0.0009992122650146484}, "greedy+LS": {"
obj": 205.0, "time": 0.07894468307495117}, "ilp": {"obj": 210.0, "status": "Time limit reached. (HiGHS St
atus 13: Time limit reached)", "runtime": 60.06559681892395, "gap": 0.3809523809523801, "nodes": 207, "bo
und": 130.0}}
{"n": 40, "T": 15, "edges": 427, "random": {"mean": 482.7000000000001, "std": 23.930942313247925, "feasib
le_runs": 10, "time": 0.010338497161865235}, "greedy": {"obj": 411.6, "time": 0.0017080307006835938}, "gr
eedy+LS": {"obj": 299.6, "time": 0.4405035972595215}, "ilp": {"obj": null, "status": "Time limit reached.
 (HiGHS Status 13: model_status is Time limit reached; primal_status is None)", "runtime": 60.56762480735
779, "gap": null, "nodes": null, "bound": null}}
{"n": 60, "T": 15, "edges": 735, "random": {"mean": 640.3, "std": 15.900943368240792, "feasible_runs": 10
, "time": 0.20722265243530275}, "greedy": {"obj": 638.9, "time": 0.0038836002349853516}, "greedy+LS": {"o
bj": 481.9, "time": 0.9924004077911377}, "ilp": {"obj": null, "status": "Time limit reached. (HiGHS Statu
s 13: model_status is Time limit reached; primal_status is None)", "runtime": 60.2475426197052, "gap": nu
ll, "nodes": null, "bound": null}}
```

```
PS C:\Users\niranjan patil\Desktop\sem5\ot\241070055_Niranjan_Patil__OT_lab8\optimize-anything> cd trackF
PS C:\Users\niranjan patil\Desktop\sem5\ot\241070055_Niranjan_Patil__OT_lab8\optimize-anything\trackF> python run.py
=== Track F — Compiler Optimization ===
--> Benchmarking Program1_Expressions...
    Code size: 18 -> 1 instrs (-94.44%)
    Dynamic cycles: 18 -> 1 (-94.44%)
    Verification: PASS (Outputs identical: [45])

--> Benchmarking Program2_Loop_Invariants...
    Code size: 28 -> 11 instrs (-60.71%)
    Dynamic cycles: 73 -> 29 (-60.27%)
    Verification: PASS (Outputs identical: [8])

--> Benchmarking Program3_Register_Pressure...
    Code size: 23 -> 1 instrs (-95.65%)
    Dynamic cycles: 23 -> 1 (-95.65%)
    Verification: PASS (Outputs identical: [21])

--> Running Register Allocation Optimization Experiment (Greedy vs ILP)...
    Total spills across 20 configurations: Greedy = 73, ILP = 51
    ILP reduced spills by 22 (30.14% reduction)

[OK] Results saved to C:\Users\niranjan patil\Desktop\sem5\ot\241070055_Niranjan_Patil__OT_lab8\optimize-anything\trackF\..\results\trackF_results
.json
PS C:\Users\niranjan patil\Desktop\sem5\ot\241070055_Niranjan_Patil__OT_lab8\optimize-anything\trackF>
```

```
PS C:\Users\niranjan patil\Desktop\sem5\ot\241070055_Niranjan_Patil__OT_lab8\optimize-anything> python -c "import json;print(json.dumps(json.load(
open('results/trackG_results.json'))['aggregate'],indent=1))"
{
 "rr": {
  "fair8_jain_cputicks": {
   "mean": 0.9984025559105432,
   "std": 0.0,
   "n": 10
  },
  "fair8_total_iters": {
   "mean": 120864.0,
   "std": 19048.275024264007,
   "n": 10
  },
  "forkstorm_total_forks": {
   "mean": 11259.7,
   "std": 1825.2883635195838,
   "n": 10
  },
  "pingpong_hog_iters": {
   "mean": 115381.9,
   "std": 28156.94006262044,
   "n": 10
  },
  "pingpong_roundtrips": {
   "mean": 24.0,
   "std": 0.0,
   "n": 10
  },
  "prop_L1_error_vs_tickets": {
   "mean": 0.32666666666666666,
   "std": 0.0,
   "n": 10
  },
```

```
PS C:\Users\niranjan patil\Desktop\sem5\ot\241070055_Niranjan_Patil__OT_lab8\optimize-anything\trackH> python run.py
=== Track H — DBMS Optimization ===
--> Generating synthetic dataset (100,000 rows)...
    Dataset created: 3.48 MB on disk

--> Evaluating Baseline (Default Optimizer, 0 secondary indexes)...
    Baseline latency: 131.53 ± 12.85 ms

--> Profiling 8 candidate indexes...
    idx_cust_acct       : Size= 160.0 KB, Benefit= 10.54 ms, Ratio=0.0659
    idx_cust_mkt        : Size= 172.0 KB, Benefit= 20.62 ms, Ratio=0.1199
    idx_orders_date     : Size= 532.0 KB, Benefit= 18.74 ms, Ratio=0.0352
    idx_orders_cust     : Size= 532.0 KB, Benefit= 19.84 ms, Ratio=0.0373
    idx_lineitem_order  : Size= 620.0 KB, Benefit= 39.09 ms, Ratio=0.0631
    idx_lineitem_price  : Size= 976.0 KB, Benefit=  6.97 ms, Ratio=0.0071
    idx_orders_price    : Size= 976.0 KB, Benefit= 20.83 ms, Ratio=0.0213
    idx_lineitem_disc   : Size= 976.0 KB, Benefit=  9.35 ms, Ratio=0.0096

--> Setting Storage Budget B = 2224.8 KB (45% of all candidates)

    Greedy Index Selection: ['idx_cust_mkt', 'idx_cust_acct', 'idx_lineitem_order', 'idx_orders_cust', 'idx_orders_date'] (Predicted Benefit: 108.
83 ms, Cost: 2016.0 KB)
    DP Index Selection:     ['idx_cust_acct', 'idx_cust_mkt', 'idx_orders_date', 'idx_orders_cust', 'idx_lineitem_order'] (Predicted Benefit: 108.
83 ms, Cost: 2016.0 KB)
    ILP Index Selection:    ['idx_cust_acct', 'idx_cust_mkt', 'idx_orders_date', 'idx_orders_cust', 'idx_lineitem_order'] (Predicted Benefit: 108.
83 ms, Cost: 2016.0 KB)

[OK] Correctness verified: All 5 query results are strictly identical before and after index optimization!
    Optimized latency: 81.50 ± 4.72 ms
    Speedup: 1.61x (38.04% latency reduction)

[OK] Results saved to C:\Users\niranjan patil\Desktop\sem5\ot\241070055_Niranjan_Patil__OT_lab8\optimize-anything\trackH\..\results\trackH_results
.json
PS C:\Users\niranjan patil\Desktop\sem5\ot\241070055_Niranjan_Patil__OT_lab8\optimize-anything\trackH>
```

```
PS C:\Users\niranjan patil\Desktop\sem5\ot\241070055_Niranjan_Patil__OT_lab8\optimize-anything> cd trackI
PS C:\Users\niranjan patil\Desktop\sem5\ot\241070055_Niranjan_Patil__OT_lab8\optimize-anything\trackI> python run.py
=== Track I — Quantum Algorithms & QML Advantage ===
STATUS: SIMULATED on classical hardware (default.qubit / default.mixed)

--> Running I1: Deutsch-Jozsa...
--> Running I2: Bernstein-Vazirani...
--> Running I3: Grover vs Linear Search (Query Count AND Time)...
--> Running I4: Quantum Fourier Transform (QFT)...
--> Running I5: Quantum Phase Estimation (QPE)...
--> Running I6: Shor's Algorithm...
--> Running I7: Simulated Quantum Annealing...
--> Running I8: Hardware Noise Model...
--> Running I9: Quantum Kernel vs Classical RBF SVM...

[OK] Results saved to C:\Users\niranjan patil\Desktop\sem5\ot\241070055_Niranjan_Patil__OT_lab8\optimize-anything\trackI\..\results\trackI_results
.json
PS C:\Users\niranjan patil\Desktop\sem5\ot\241070055_Niranjan_Patil__OT_lab8\optimize-anything\trackI>
```

```
PS C:\Users\niranjan patil\Desktop\sem5\ot\241070055_Niranjan_Patil__OT_lab8\optimize-anything\trackJ> python profile_suite.py
LOC=3191 tests=85 functions covered=111 suite_s=22.17
PS C:\Users\niranjan patil\Desktop\sem5\ot\241070055_Niranjan_Patil__OT_lab8\optimize-anything\trackJ> python run.py
=== Track J — Software Engineering Optimization (measured) ===
codebase: 3191 source lines; 85 tests; 111 functions covered; sum(test time)=22.17s
{
 "codebase": {
  "name": "optimize-anything (this repo)",
  "source_lines_track_dirs": 3191,
  "tests_profiled": 85,
  "functions_covered_by_full_suite": 111,
  "sum_of_test_times_s": 22.169,
  "full_suite_wall_s": 32.589,
  "full_suite_wall_runs_s": [
   33.565,
   32.589,
   31.531
  ]
 },
 "problem_a_minimization": {
  "naive_full_suite": {
   "tests": 85,
   "predicted_s": 22.169,
   "wall_s": 32.589,
   "coverage": 1.0
  },
  "greedy_set_cover": {
   "tests": 46,
   "predicted_s": 15.008,
   "wall_s": 32.553,
   "wall_runs_s": [
    33.383,
```

```
PS C:\Users\niranjan patil\Desktop\sem5\ot\241070055_Niranjan_Patil__OT_lab8\optimize-anything\trackJ> cd..
PS C:\Users\niranjan patil\Desktop\sem5\ot\241070055_Niranjan_Patil__OT_lab8\optimize-anything> cd compare
PS C:\Users\niranjan patil\Desktop\sem5\ot\241070055_Niranjan_Patil__OT_lab8\optimize-anything\compare> python  run_comparison.py
{
 "knapsack_instances": 40,
 "maxcut_graphs": 20,
 "knapsack": {
  "claude_chat": {
   "greedy": {
    "infeasible": 0,
    "optimal": 26,
    "mean_ms": 0.05251254187896848
   },
   "dp": {
    "infeasible": 0,
    "optimal": 40,
    "mean_ms": 0.0897524761967361
   },
   "bnb": {
    "infeasible": 0,
    "optimal": 40,
    "mean_ms": 0.0984274665825069
   },
   "ilp": {
    "infeasible": 0,
    "optimal": 40,
    "mean_ms": 2.325027482584119
   }
  },
  "antigravity": {
```

```
PS C:\Users\niranjan patil\Desktop\sem5\ot\241070055_Niranjan_Patil__OT_lab8\optimize-anything> python -m pytest tests -q
=== B1. 0/1 Knapsack ===
  Greedy=147  DP=151  B&B=151  ILP=151
  Times: greedy=0.0001s  dp=0.0002s  bnb=0.0001s  ilp=0.0057s

=== B2. Weighted Interval Scheduling ===
  Greedy-EFT weight=25  DP weight=23  gap=-8.7%

=== B3. Max-Cut (7 nodes) ===
  Edges=11  Greedy=22  Brute-force=33
  SA 10 runs: mean=33.0  std=0.0

=== B4. QUBO / Ising formulation ===
  QUBO unconstrained: brute-force energy=-33.00  cut=33
  SA unconstrained 10 runs: mean=-33.00  std=0.00
  QUBO constrained (|S|=3, lam=10.0): brute-force energy=-123.00  cut=33
  Ising(h,J)+const = -33.0000  QUBO energy = -33.0000  (match OK)
.....................................................................s.....................                                [100%]
=================================================== warnings summary ===================================================
tests/test_trackC.py::TestTrackCFiniteShotsAndNoise::test_shot_noise_variance_decreases
  C:\Users\niranjan patil\anaconda3\Lib\site-packages\pennylane\devices\device_api.py:207: PennyLaneDeprecationWarning: Setting sh
ots on device is deprecated. Please use the `set_shots` transform on the respective QNode instead.
    warnings.warn(

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
88 passed, 1 skipped, 1 warning in 22.96s
PS C:\Users\niranjan patil\Desktop\sem5\ot\241070055_Niranjan_Patil__OT_lab8\optimize-anything>
```